Beauty Meson Decay Analysis: Background Reduction and Resonance Study in $B_s^0 \to K^+\pi^-K^-\pi^+$

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I. Introduction

THE decay $B_s^0 \to K^+\pi^-K^-\pi^+$ is a charmless four-body hadronic decay proceeding via a $b \to s\bar{d}d$ flavour-changing neutral current transition, mediated at the quark level by a QCD penguin loop, as shown in Fig. 1. Tree-level contributions are strongly suppressed by the small CKM matrix element $|V_{ub}|$, making this a penguin-dominated process [1], [2].Crucially, the underlying $b \to s\bar{d}d$ transition is distinct from the more commonly studied $b \to s\bar{s}s$ penguin modes. This allows access to a complementary CP-violating phase, $\phi_s^{d\bar{d}}$, which had not been measured prior to the LHCb Run 1 analysis. The measured value, $\phi_s^{d\bar{d}} = -0.10 \pm 0.13\,(\text{stat}) \pm 0.14\,(\text{syst})$ rad [3], is consistent with the Standard Model but remains statistically limited, motivating further study.

Beyond CP violation, this decay presents a striking challenge to theory through the polarisation puzzle. The dominant quasi-two-body mode $B_s^0 \to K^*(892)^0\bar{K}^*(892)^0$ is measured to have a longitudinal polarisation fraction $f_L = 0.240 \pm 0.040$ [1], far below the $f_L \approx 1$ predicted by naïve QCD factorisation [4], [5]. This is in stark contrast to the U-spin related mode $B_d^0 \to K^*(892)^0\bar{K}^*(892)^0$, where $f_L = 0.724 \pm 0.051$ [1], a factor of three larger despite the two decays differing only by the spectator quark flavour. No satisfactory Standard Model explanation currently exists for this discrepancy [2], [5].

The four-body final state is particularly valuable as it simultaneously provides access to two $K\pi$ invariant mass spectra, three helicity angles, and decay-time information, enabling a full amplitude analysis to disentangle resonant and non-resonant contributions. Since new physics particles can enter the penguin loop and modify the decay amplitudes, branching fractions and CP phases [6], [7], precision measurements of this channel offer a powerful probe of physics beyond the Standard Model.

II. Reducing Combinatorial Background

To train a classifier to distinguish signal from combinatorial background, the signal sample is taken from simulated $B_s^0 \to K^+\pi^-K^-\pi^+$ decays (`mc.root`), while the background sample is drawn from the upper mass sideband ($m(B_s^0) > 5600$ MeV/c^2) of the reconstructed data collected by the LHCb detector (`data.root`).This threshold lies well above both the B_s^0 mass and the B^0 mass (5279.66 ± 0.12 MeV/c^2 [8]), ensuring that the sideband is free from signal and peaking backgrounds from other B meson decays, and thus provides a high-purity sample of combinatorial background.

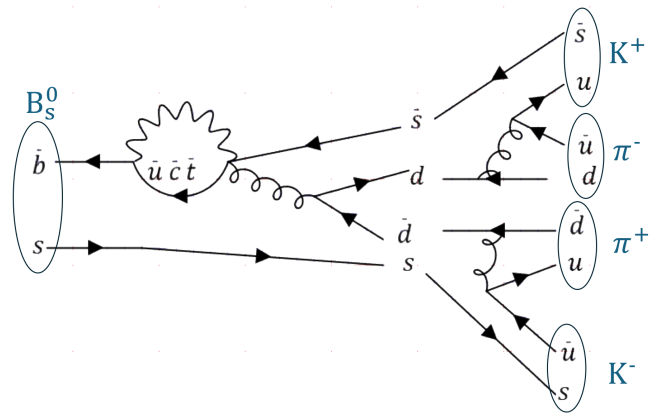

Fig. 1. Leading-order QCD penguin diagram contributing to $B_s^0 \to K^+\pi^-K^-\pi^+$. The $\bar{b}$ quark undergoes a flavour-changing neutral current transition $\bar{b} \to \bar{d}d\bar{s}$ via a loop containing virtual up-type quarks (u, c, t). The emitted gluon produces a $d\bar{d}$ pair, while the spectator s quark remains unchanged. The quark system subsequently hadronises through non-perturbative QCD processes, including the creation of additional $u\bar{u}$ pairs, forming the four-body final state $K^+(u\bar{s})\,\pi^-(d\bar{u})\,K^-(\bar{u}s)\,\pi^+(u\bar{d})$.

A. Feature Selection

Since combinatorial background consists of combinations of genuine, correctly reconstructed $K^\pm$ and $\pi^\pm$ tracks, it is primarily distinguished from signal by the absence of a common displaced decay topology rather than by individual track properties. Single-particle variables for $K^\pm$ and $\pi^\pm$—including PID response, individual momenta, and track-quality observables—are therefore expected to provide limited discrimination and were not included as input features.

The reconstructed B_s^0 mass and all kinematically correlated quantities—mass-constrained K^* masses (`Bs_Kst_M`, `Bs_Kstb_M`) and unconstrained invariant masses (`Kst_M`, `Kstb_M`)—were excluded to prevent sculpting of the background mass distribution. Cartesian four-momentum components (`_PX`, `_PY`, `_PZ`, `_E`) were replaced by scalar quantities such as `Bs_PT` and `Bs_P`, removing detector-coordinate dependence without loss of discrimination. Absolute vertex positions and non-physical variables (`Year`, `MagnetPolarity`, PDG identifiers, fixed masses) were also removed.

After these exclusions, 23 candidate features remain, describing the topological and kinematic properties of the B_s^0 and K^* candidates. A two-sample Kolmogorov–Smirnov test was used to rank their separation power (Fig. 3). The five most discriminating variables all reflect the displaced decay

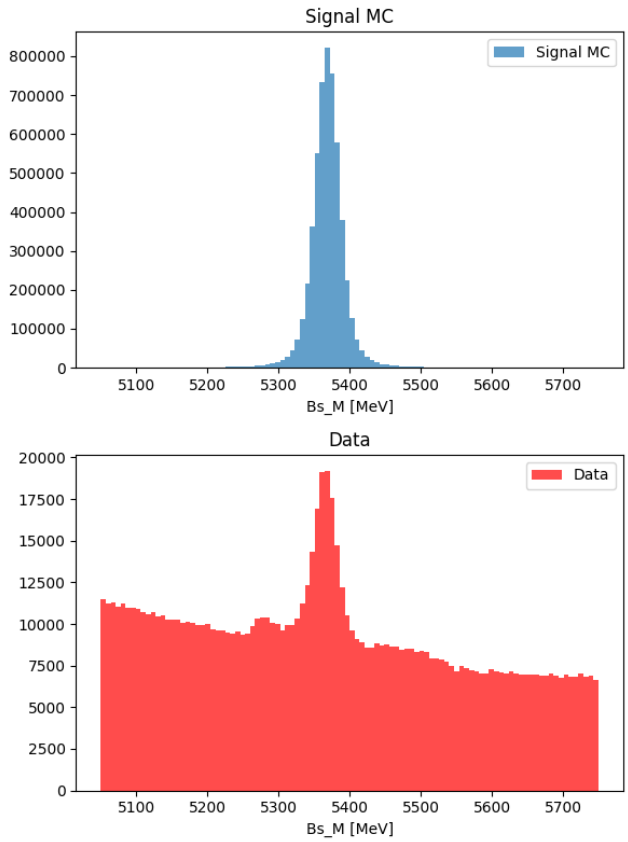

Fig. 2. Invariant mass distribution of the $K^+\pi^-K^-\pi^+$ system for signal simulation (left) and data (right). The simulated signal exhibits a narrow peak around the B_s^0 mass, reflecting the detector resolution. In contrast, the data sample shows a prominent signal peak on top of the background. The high-mass region ($m(B_s^0) > 5600$ MeV/c^2) is approximately structureless, consistent with being dominated by combinatorial background, and is therefore used to define a background training sample.

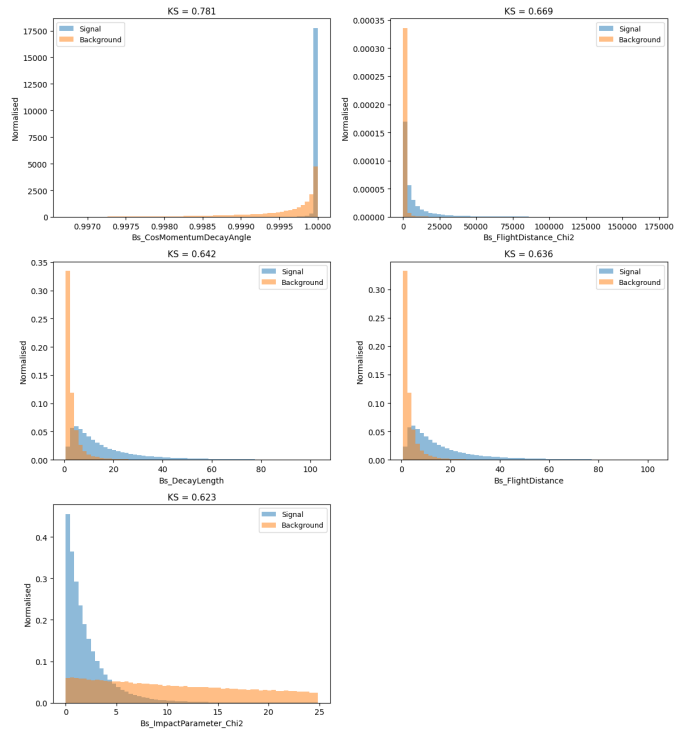

Fig. 3. Normalised distributions of the five most discriminating features, ranked by two-sample Kolmogorov–Smirnov statistic, for signal (blue) and combinatorial background (orange). `Bs_CosMomentumDecayAngle` peaks sharply near unity for signal, reflecting the alignment of the B_s^0 momentum with its flight direction, while background is broadly distributed. `Bs_FlightDistance_Chi2`, `Bs_DecayLength`, and `Bs_FlightDistance` all show signal extending to large values due to the measurable B_s^0 flight path, whereas background clusters near zero where no genuine displacement exists. `Bs_ImpactParameter_Chi2` exhibits the opposit behaviour: signal is concentrated at small values since a genuine B_s^0 points back to the primary vertex, while background populates a broader range.

topology of a genuine B_s^0 meson. Its finite lifetime ($\tau \approx 1.5$ ps [8]) produces measurable flight distances and large flight-distance significances absent in random track combinations. Furthermore, since the B_s^0 originates at the primary vertex, its momentum vector points back to it, yielding a small impact parameter and a direction-angle cosine close to unity—geometric constraints that combinatorial candidates do not satisfy.

B. Training

With the 23 selected features, the signal Monte Carlo sample was downsampled to match the background sideband size (∼0.15M), yielding balanced training classes. The combined dataset was randomly split into training (60%), validation (20%), and test (20%) subsets. A gradient-boosted decision tree (XGBoost) was trained using binary cross-entropy (log loss) as the objective function, with the simulation `weight` branch applied as sample weights to correct for MC–data discrepancies. A grid search was performed over three hyperparameters: the number of boosting rounds (`n_estimators` $\in \{100, 200, 300, 400\}$), tree depth (`max_depth` $\in \{2, 3, 4\}$), and learning rate (`learning_rate` $\in \{0.01, 0.05, 0.1\}$), while the row subsampling fraction was fixed at 0.8 to provide a baseline level of regularisation without introducing an additional search dimension. Each configuration was evaluated using the area under the ROC curve (AUC) on the validation set, which provides a threshold-independent measure of classifier performance. The train–validation AUC gap was simultaneously monitored as a proxy for overtraining (Fig. 4). The final configuration was selected by requiring a gap below 0.002 and maximising the validation AUC (0.995), yielding `n_estimators` = 300, `max_depth` = 4, `learning_rate` = 0.1, and `subsample` = 0.8.

The training curve (Fig. 5) shows both training and test loss decreasing throughout without divergence. The classifier output distributions for training and test samples are compared in Fig. 6, with Kolmogorov–Smirnov statistics of 0.007 for both the signal and background classes and p-values of 0.30 and 0.24 respectively, confirming no evidence of overtraining.

C. Determining the Cut Point

To determine the optimal BDT selection threshold, we follow a figure-of-merit (FOM) approach. First, the BDT is evaluated on the labelled training sample (signal MC and data sideband) to obtain the signal efficiency $\varepsilon_S(t)$ and background efficiency $\varepsilon_B(t)$ as functions of the cut threshold t, shown

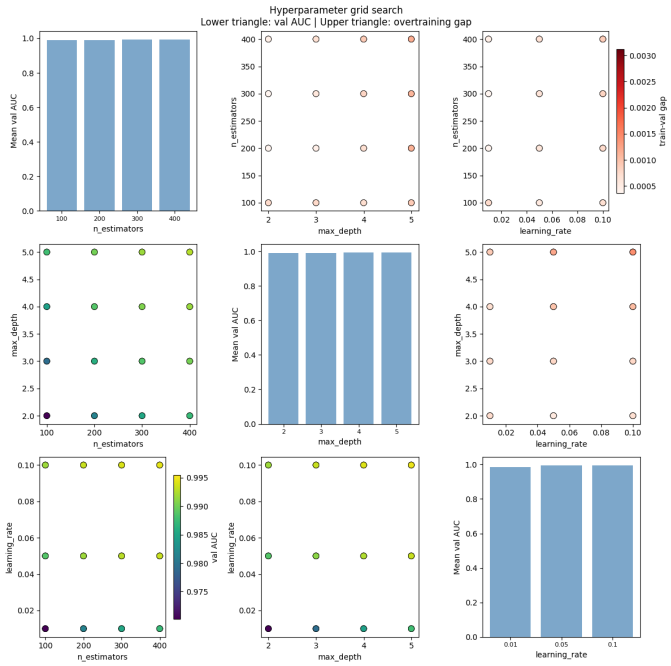

Fig. 4. Hyperparameter grid search results with `subsample` fixed at 0.8. Diagonal panels show the mean validation AUC for each parameter value, marginalised over the other two. Lower-triangle panels show pairwise scatter plots coloured by validation AUC (brighter is higher), while upper-triangle panels show the same configurations coloured by the train–validation AUC gap (darker indicates more overtraining).

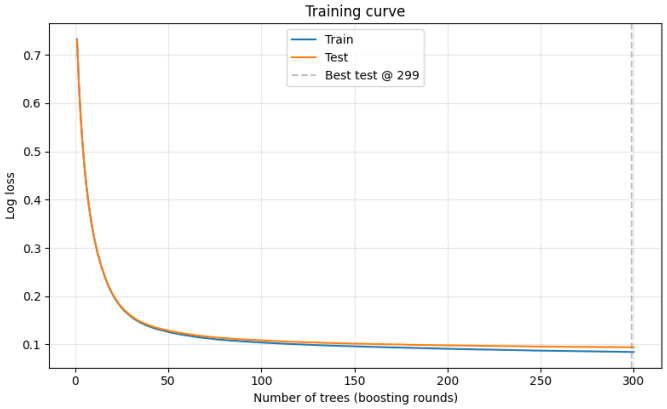

Fig. 5. Training and test log loss as a function of boosting round. Both curves decrease monotonically with a small, stable gap, indicating no overtraining.

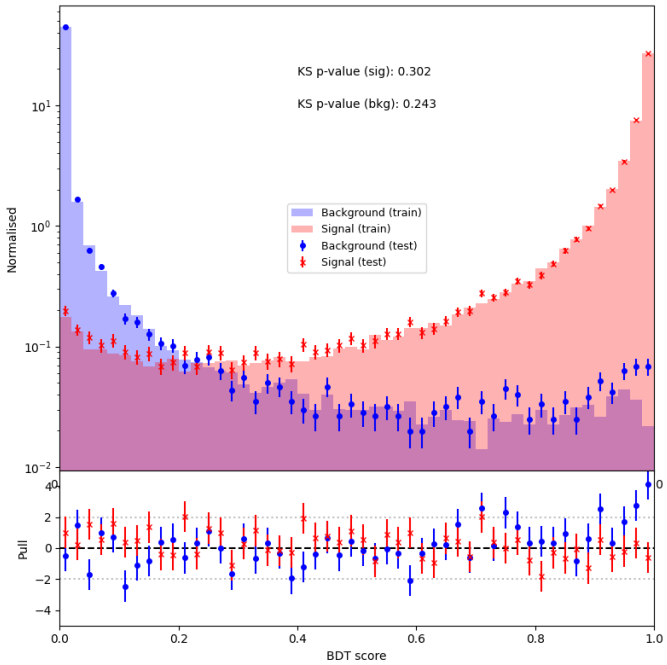

Fig. 6. Upper panel: normalised BDT output score distributions for signal (red) and background (blue), comparing training samples (filled histograms) with independent test samples (points with error bars). Good agreement is observed across the full score range for both classes, with KS p-values of 0.30 (signal) and 0.24 (background). Lower panel: bin-by-bin pull, defined as (test − train)/σ, where σ is the combined statistical uncertainty. The pulls are consistent with zero and in general remain within $\pm 2\sigma$.

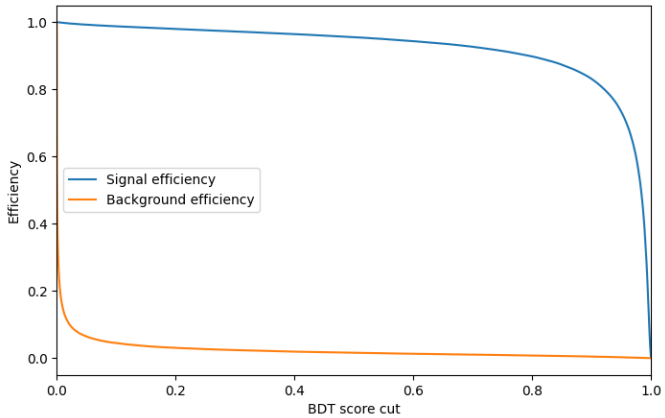

Fig. 7. Signal and background efficiencies as functions of the BDT score cut threshold.The background efficiency drops sharply at low BDT scores, while the signal efficiency remains high over a wide range before falling, reflecting the good discriminating power of the classifier.

in Fig. 7. An extended binned maximum-likelihood fit to the B_s invariant mass distribution over the full range is performed using a simplified model, with the signal described by a truncated Gaussian and the combinatorial background by a truncated exponential. This provides initial estimates of the signal and background yields, $N_S^0 = 61.8 \pm 0.5k$ and $N_B^0 = 877.5 \pm 1.0k$. The pull distribution (Fig. 8) shows no significant bias across the mass range, indicating that the model adequately captures the overall shape. The expected yields after a cut at threshold t are then

$$S(t) = N_S^0 \varepsilon_S(t), \qquad B(t) = N_B^0 \varepsilon_B(t), \tag{1}$$

from which the significance is computed as

$$\mathrm{FOM}(t) = \frac{S(t)}{\sqrt{S(t) + B(t)}}. \tag{2}$$

The optimal cut (0.779, with corresponding $\frac{S}{\sqrt{S+B}} = 222.8$) is chosen at the value of t that maximises the FOM, which keeps 90.5% of the signal and rejects 99.2% of the background. as shown in Fig. 9. Figure 10 shows the B_s mass distribution before and after the optimal BDT cut is applied, demonstrating a substantial reduction in combinatorial background.

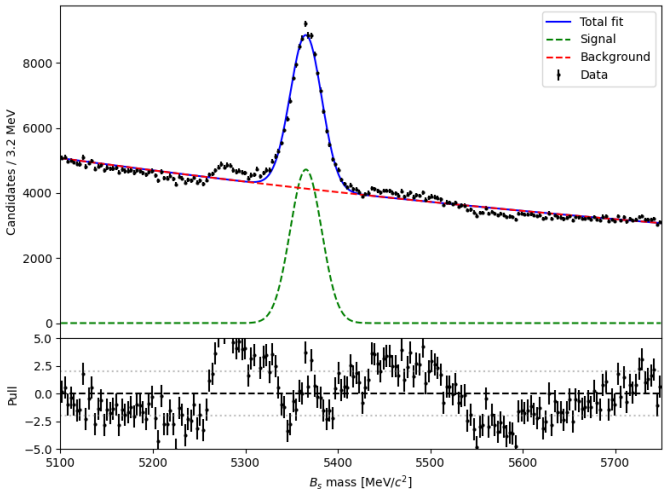

Fig. 8. Extended binned maximum-likelihood fit to the B_s invariant mass distribution of the data sample. The lower panel shows the pull distribution, $(N_{\text{obs}} - N_{\text{exp}})/\sqrt{N_{\text{exp}}}$.The localised excesses, most notably near $5280\text{MeV}/c^2$, are consistent with known peaking backgrounds and constitute only a small fraction of the total background yield, so their impact on the FOM curve and the resulting optimal cut is negligible.

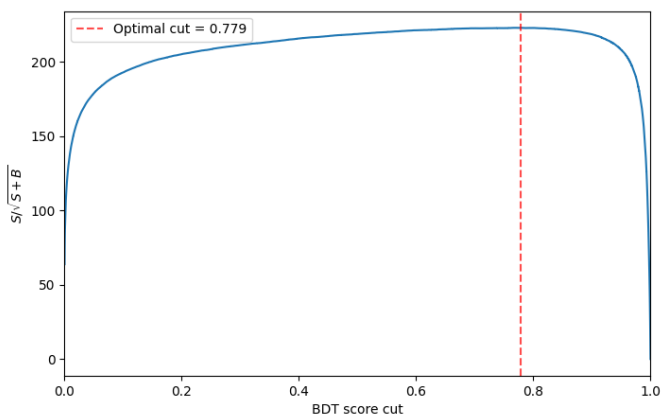

Fig. 9. Figure of merit $S/\sqrt{S+B}$ as a function of the BDT score cut threshold.

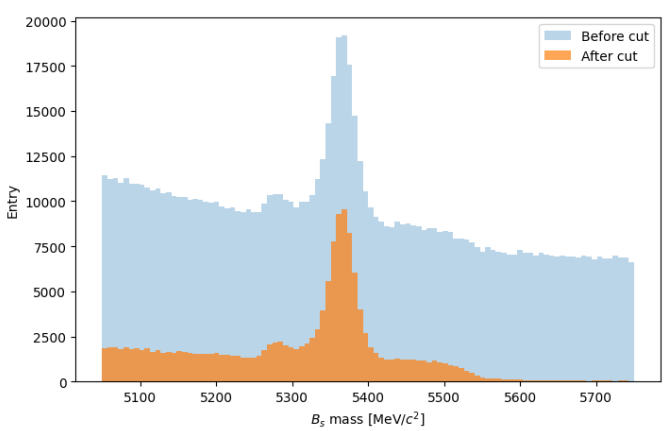

Fig. 10. B_s invariant mass distribution before (blue) and after (orange) the optimal BDT selection. The cut significantly suppresses the combinatorial background while preserving the signal peak near $5367\text{MeV}/c^2$.

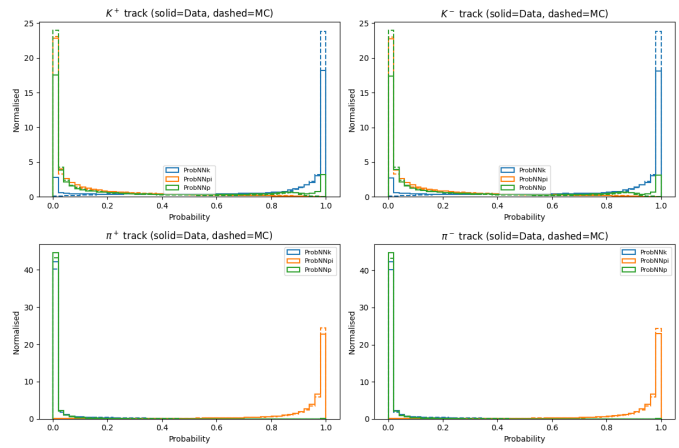

Fig. 11. PID probability distributions for each of the four tracks, comparing signal MC (dashed) and BDT-selected data (solid).

III. Reducing Other Backgrounds

Having removed 99.2% combinatorial background, the remaining sample is dominated by candidates in which all four tracks originate from a common displaced vertex, which can be categorised as follows: (i) misidentified decays, where a genuine parent particle decay to a different final state enters the sample due to incorrect particle identification; (ii) $B^0 \to K^+\pi^-K^-\pi^+$, which shares the same final state as the signal; and (iii) partially reconstructed decays with a missing neutral particle, which populate the low-mass region below the signal peak. In this section we address category (i) using particle identification requirements.

The LHCb particle identification system provides neural-network-based probabilities for each track to be a kaon (ProbNNk), pion (ProbNNpi), or proton (ProbNNp). Figure 11 shows these distributions for signal MC and BDT-selected data. For kaon candidates, data exhibit an excess at low ProbNNk and high ProbNNp, indicating contamination from $\Lambda_b \to p\pi^-K^-\pi^+$ decays where a proton is misidentified as a kaon. In contrast, pion distributions agree well between data and MC, suggesting that $K \leftrightarrow \pi$ and $p \leftrightarrow \pi$ misidentification is small and that additional pion PID cuts would provide limited benefit.

To confirm the source of the contamination, the four-body invariant mass is recomputed under alternative mass hypotheses (Fig. 12). Assigning the proton mass to the K^+ track produces a clear peak at $5620;\text{MeV}/c^2$. No analogous structure is observed for other swaps.

The proton veto threshold is optimised using the Λ_b peak at $5620\text{MeV}/c^2$ under the $K^+ \to p$ mass hypothesis (Fig. 13). A requirement of ProbNNp < 0.5, applied to both K^+ and K^- candidates, removes the peak while retaining about 75% of the signal MC. Although this reduces signal efficiency, it suppresses $\Lambda_b \to p\pi^-K^-\pi^+$ contamination at the selection stage, avoiding the need for an additional fit component and improving the reliability of subsequent sWeight-based projections. As shown in Fig. 14, the broad excess in the 5400–$5550\text{MeV}/c^2$ region is effectively eliminated.

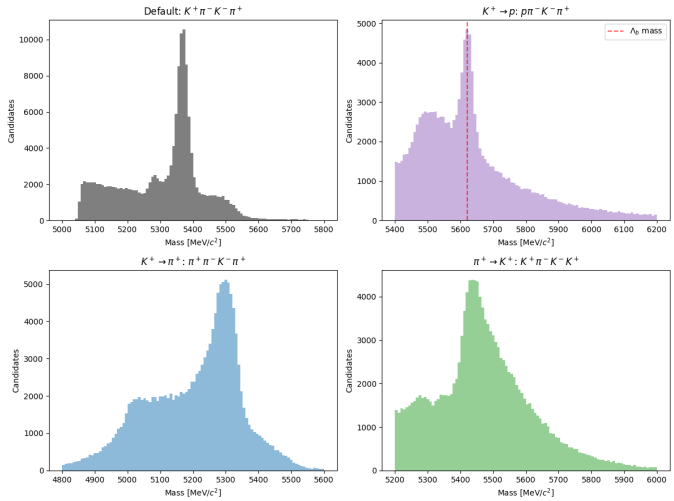

Fig. 12. Four-bodyinvariantmass of BDT-selecteddatarecomputed under alternative mass hypotheses to identify peaking backgrounds. **Top left:** default $K^+\pi^-K^-\pi^+$ assignment.**Top right:** $K^+ \to p$ substitution, testing for $\Lambda_b \to p\pi^-K^-\pi^+$ contamination (dashed line marks the Λ_b mass at $5620\,\mathrm{MeV}/c^2$). **Bottom left:** $K^+ \to \pi^+$ substitution, testing for $B^0 \to \pi^+\pi^-K^-\pi^+$. **Bottom right:** $\pi^+ \to K^+$ substitution, testing for $B_s \to K^+\pi^-K^-K^+$.

IV. B Candidate Mass Fit

An extended binned maximum-likelihood fit is performed on the B_s candidate mass spectrum after the BDT and proton veto selections using 500 bins. The total model is

$$f(m) = N_s \cdot f_{\mathrm{sig}}(m) + N_0 \cdot f_{B^0}(m) + N_b \cdot f_{\mathrm{comb}}(m) + N_p \cdot f_{\mathrm{part}}(m). \tag{3}$$

The four components are summarised in Table I.

The key parameters, listed with their fitted values and uncertainties in Table II. Deviations from unity are expected given the large data sample and residual localised structures not fully captured by the model, such as the approximate description of the signal resolution using a Crystal Ball plus Gaussian function, the finite flexibility of the Bernstein parametrisation for the combinatorial background, and possible residual misidentified backgrounds not fully removed. The pull distribution (Fig. 15) is largely contained within $\pm 2\sigma$ with no significant systematic trend, supporting the overall fit quality.

V. Projection of the Signal Component in Intermediate Masses

The signal component is projected onto intermediate mass spectra using the sPlot technique [12]. For each event i with reconstructed mass m_i, a signal weight is computed as

$$w_s(m_i) = \frac{\sum_j V_{sj} f_j(m_i)}{\sum_k N_k f_k(m_i)}, \tag{4}$$

where f_j are the PDFs of each component evaluated at m_i, N_k are the fitted yields, and V_{sj} are elements of the covariance matrix obtained from the fit. The covariance matrix is defined as

$$V_{jk}^{-1} = \sum_i \frac{f_j(m_i)\, f_k(m_i)}{\left(\sum_l N_l f_l(m_i)\right)^2}. \tag{5}$$

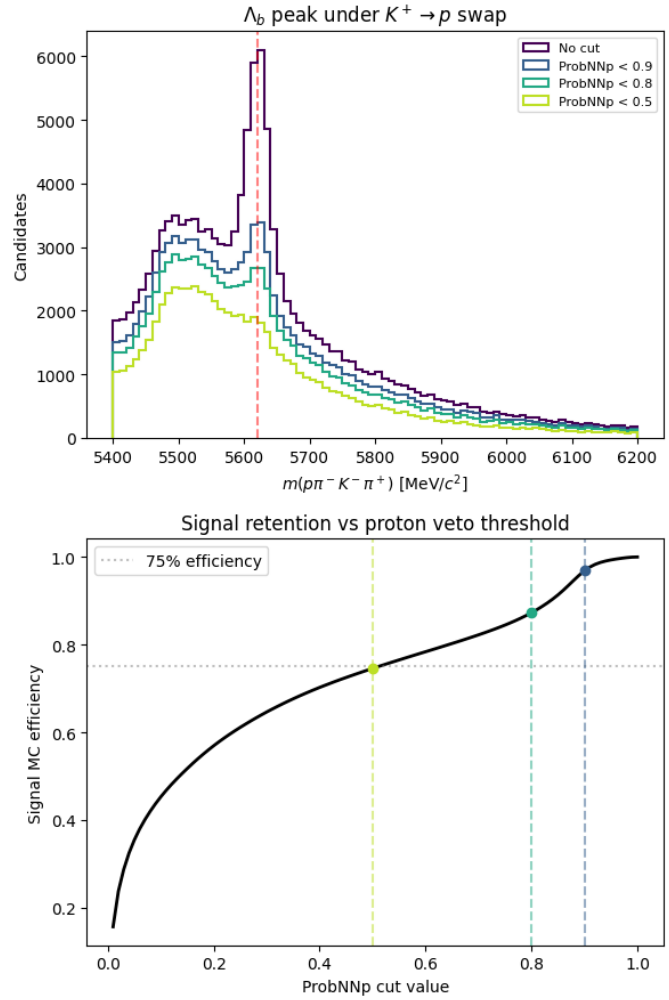

Fig. 13. Optimisation of the proton veto threshold. Up: the four-body invariant mass under the $K^+ \to p$ hypothesis for different ProbNNp cut values, showing the Λ_b peak at $5620\,\mathrm{MeV}/c^2$ progressively suppressed as the cut is tightened. Bottom: signal MC efficiency as a function of the cut value. The chosen working point, K±ProbNNp < 0.5, effectively eliminates the Λ_b peak while retaining approximately 75% of the signal.

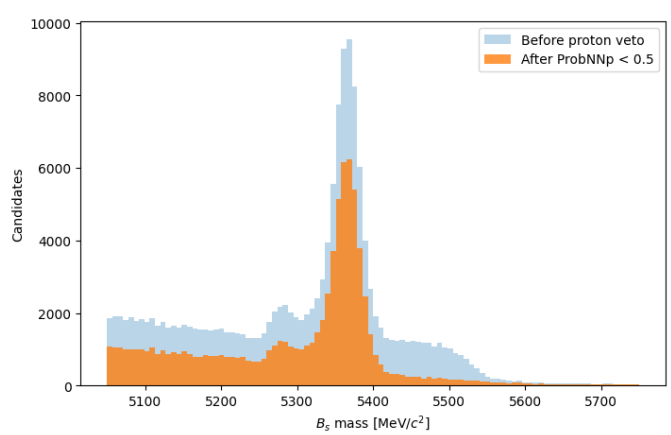

Fig. 14. B_s invariant mass distribution before (blue) and after (orange) the proton veto (K±_ProbNNp < 0.5).

When applied to other observables, these weights statistically subtract the background contribution, allowing the reconstruc-

TABLE I
COMPONENTS OF THE B-CANDIDATE MASS FIT MODEL.

Component	Model	Motivation
B_s signal	Crystal Ball + Gaussian: $f_{CB}\,\mathrm{CB}(m;\mu_s,\sigma_1,\beta,n)+(1-f_{CB})\,\mathrm{Gauss}(m;\mu_s,\sigma_2)$	The Crystal Ball core with a power-law tail accounts for the detector mass resolution and radiative losses; the additional Gaussian describes a broader resolution component, following Refs. [9].
$B^0 \to K^+\pi^-K^-\pi^+$	Same CB + Gaussian, shared shape parameters	Identical final state to the signal; the detector resolution is therefore expected to be the same. Only the mean μ_0 differs.
Combinatorial	3rd-order Bernstein polynomial (4 coefficients b_0–b_3)	Provides a flexible, smooth parameterisation of the residual combinatorial background without assuming a specific functional form [10].
Partially reconstructed	ARGUS function [11] $\otimes$ Gaussian	Describes b-hadron decays with a missing neutral particle (e.g. π^0), following the LHCb analysis [3]. The ARGUS cutoff is fixed at $m_{B_s} - m_{\pi^0} \approx 5232\mathrm{MeV}/c^2$, corresponding to the kinematic endpoint when a π^0 carries zero momentum. The convolution with a Gaussian accounts for the detector mass resolution.

TABLE II
RESULTS OF THE B_s MASS FIT.

Parameter	Fit value	Comment
B_s^0 signal yield	40377 ± 513	
B^0 background yield	3717 ± 184	
Combinatorial background yield	36637 ± 1482	
Partially reconstructed yield	2561 ± 1075	Large uncertainty due to correlation with the flexible combinatorial background; limited impact as this component contributes only in the low-mass region
B_s^0 mass	5365.56 ± 0.15 MeV	Consistent with PDG value
B^0 mass	5280.45 ± 0.87 MeV	Consistent with PDG value
Crystal Ball fraction (f_{CB})	0.927 ± 0.009	
Core mass resolution (σ_1)	16.37 ± 0.15 MeV	Detector resolution of the main signal peak
Gaussian width (σ_2)	50.00 ± 3.63 MeV	Constrained to an upper bound to prevent the signal model from absorbing background; larger values lead to unphysical broadening inconsistent with expected detector resolution
ARGUS shape parameter (c)	27.35 ± 14.25	Controls slope of partially reconstructed background

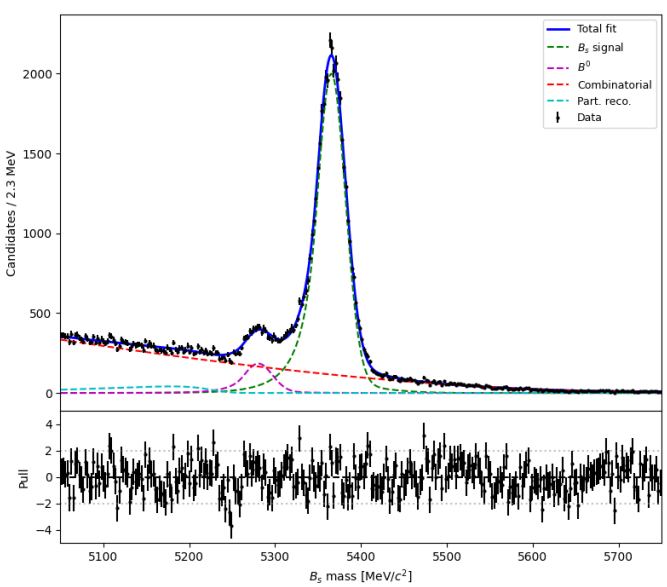

Fig. 15. Extended binned maximum-likelihood fit to the B_s candidate mass spectrum.

tion of the signal-only distribution under the assumption that the fitted variable (the B candidate mass) is uncorrelated with the variables of interest.

The intermediate invariant masses are constructed from the four-momenta of the decay products, assigning the known kaon and pion masses:

$$\begin{aligned} m(K^+\pi^-) &= \sqrt{(p_{K^+}+p_{\pi^-})^2}, \\ m(K^-\pi^+) &= \sqrt{(p_{K^-}+p_{\pi^+})^2}. \end{aligned} \tag{6}$$

The sWeighted distributions of $m(K^+\pi^-)$ and $m(K^-\pi^+)$ are shown in Fig. 16, and the Dalitz plot of $m^2(K^+\pi^-)$ vs $m^2(K^-\pi^+)$ in Fig. 17.

The $K\pi$ system can form intermediate resonant states with different total angular momentum J. As both the kaon and pion are spin-zero particles, J is determined by the orbital angular momentum L, giving rise to scalar (S-wave, $J=0$), vector (P-wave, $J=1$) and tensor (D-wave, $J=2$) contributions.

The P-wave is dominated by the narrow $K^*(892)$ resonance, producing a prominent peak around $895.5\,\mathrm{MeV}/c^2$ [8]. At higher masses, the region near $1430\,\mathrm{MeV}/c^2$ receives overlapping contributions from scalar and tensor states, notably $K_0^*(1430)$ and $K_2^*(1430)$, resulting in a broader and more complex structure.

These amplitudes interfere coherently, so the observed spectrum is not a simple sum of resonances. In addition to distinct

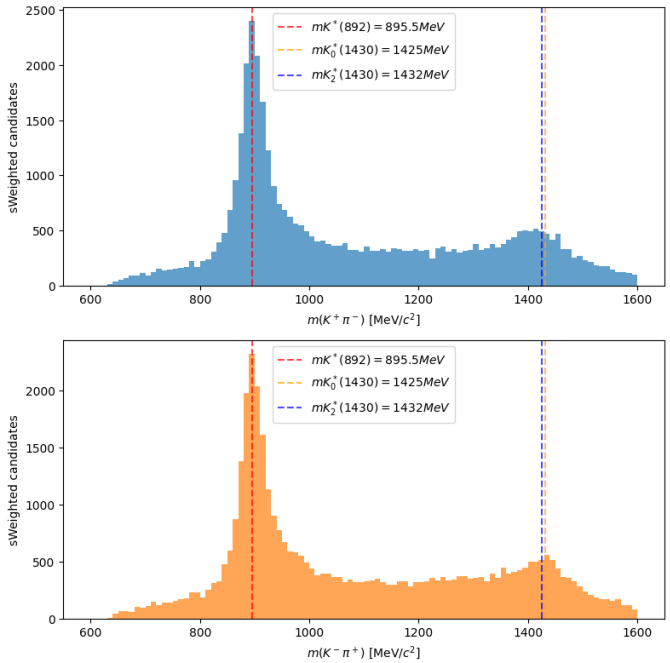

Fig. 16. sWeighted invariant mass distributions of the $K^+\pi^-$ (top) and $K^-\pi^+$ (bottom) systems. A prominent narrow peak is observed at $\sim$ 895.5 MeV/c^2, corresponding to the $K^*(892)^0$ resonance. A broader enhancement is visible around 1.4 GeV/c^2, consistent with contributions from higher-mass scalar and tensor $K\pi$ components. The smooth distribution between and beneath the peaks reflects broad S-wave contributions and interference between amplitudes.

peaks, interference and broad scalar components generate a smooth underlying distribution. The main contributions are summarised in Table III.

No significant resonant structures are observed in any of the other 2-body or 3-body invariant mass spectra (Fig. 18 and Fig. 19). The expected absence of these resonances is explained in Table IV.

VI. Conclusion

This analysis selects $B_s^0 \to K^+\pi^-K^-\pi^+$ candidates using a BDT classifier and PID vetoes, yielding 40,000 signal events from a mass fit with $\chi^2/\text{ndof} = 1.4$. The sWeighted distribution of of $m(K^+\pi^-)$ and $m(K^-\pi^+)$confirms a dominant $K^{*0}\bar{K}^0$ intermediate state. Future work should perform a full amplitude analysis to extract individual resonant fractions, relative phases, and CP-violation observables.

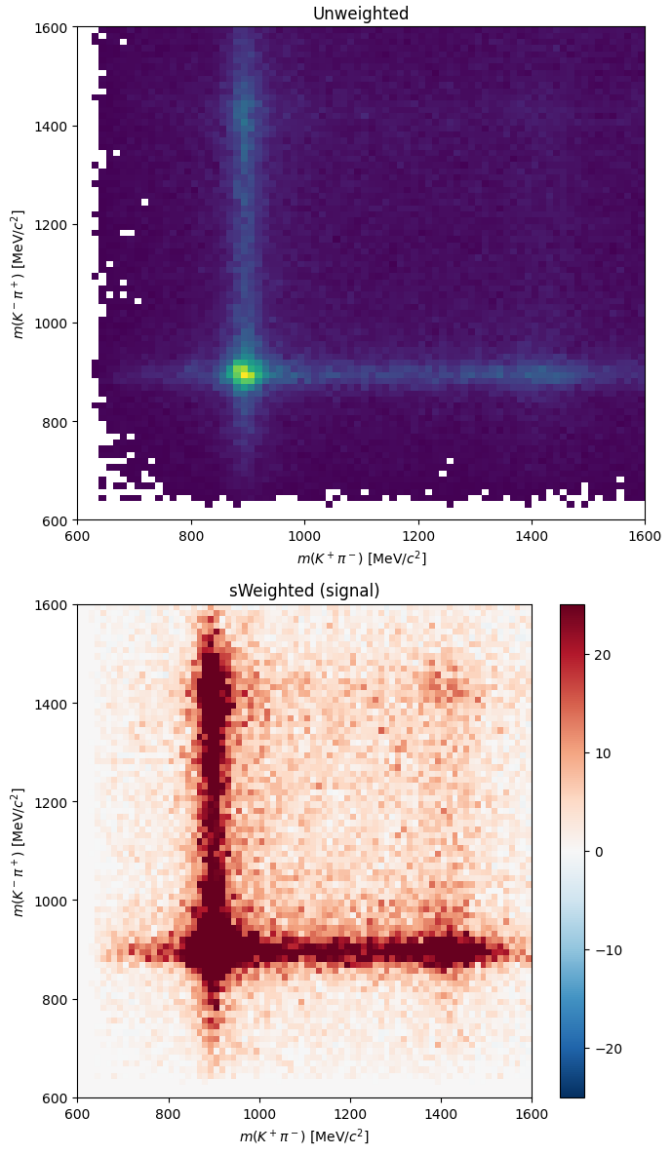

Fig. 17. Two-dimensional distributions of $m(K^+\pi^-)$ versus $m(K^-\pi^+)$ for the unweighted sample (top) and the sWeighted signal (bottom). The signal projection exhibits significantly clearer structures, with pronounced horizontal and vertical bands at $\sim$ 895.5 MeV/c^2, corresponding to $K^*(892)^0$ contributions in each $K\pi$ system.

TABLE III
MAIN CONTRIBUTIONS TO THE $K\pi$ INVARIANT-MASS SPECTRUM AND THEIR QUALITATIVE EFFECTS [3].

Type	Example contribution	Effect in $K\pi$ mass
Vector–vector	$K^*(892)^0\bar{K}^*(892)^0$	Strong narrow peak at 895.5 MeV/c^2
Scalar–vector / vector–scalar	$(K\pi)^*_0K^*(892)$	895.5 peak + broad background
Scalar–scalar	$(K\pi)^*_0(K\pi)^*_0$	Smooth non-resonant and broad S-wave contribution over a wide mass range, including contributions from scalar resonances such as $K^*_0(1430)$ and low-mass components like $K^*_0(800)$
Tensor contributions	$K^*_2(1430)$ modes	Broad enhancement near 1432 MeV/c^2
Interference	—	Distorts shapes and fills regions between peaks

TABLE IV
SUMMARY OF INVARIANT MASS COMBINATIONS AND EXPECTED RESONANT CONTRIBUTIONS IN $B^0_s \to K^+\pi^-K^-\pi^+$ DECAYS.

Combination	Possible resonance	Expected behaviour
2-body		
$m(K^+K^-)$	$\phi(1020)$	Would require $B_s \to \phi\rho$ with $\phi(s\bar{s})$ and $\rho(d\bar{d})$. In the penguin diagram, this pairing requires the s from the $b \to s$ transition to combine with the spectator $\bar{s}$, which does not contribute at leading order in the factorisation framework [2].
$m(\pi^+\pi^-)$	$\rho(770)$	Same as above
$m(K^+\pi^+)$	None	No conventional $q\bar{q}$ resonance (charge ±2)
$m(K^-\pi^-)$	None	Same as above
3-body		
$m(K^+\pi^-K^-)$	—	No natural intermediate resonance; does not correspond to a single hadron state.
$m(K^+\pi^-\pi^+)$	—	Same as above.
$m(K^+K^-\pi^+)$	$D^+_s(1968)$	$B_s \to D^\mp_s\pi^\pm$ has BR $\sim 10^{-4}$ [8], an order of magnitude above the signal. However, the D_s displaced vertex produces a distinct topology that is rejected by the BDT; no D_s peak is observed, confirming effective suppression.
$m(K^+K^-\pi^-)$	$D^-_s(1968)$	Same as above

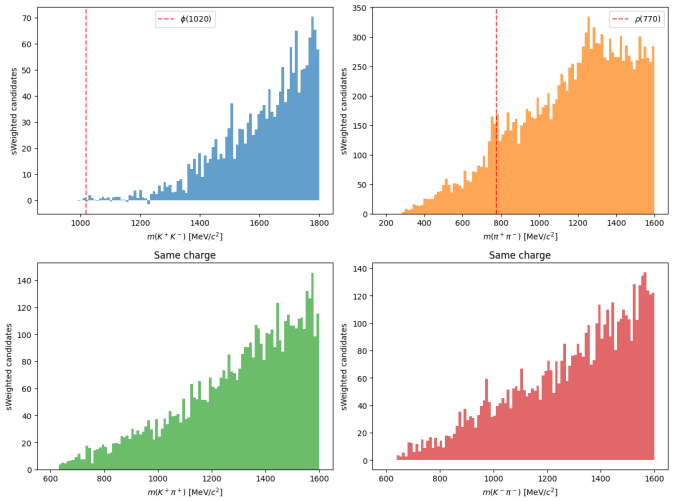

Fig. 18. sWeighted 2-body invariant mass distributions: $m(K^+K^-)$ (top left), $m(\pi^+\pi^-)$ (top right), and same-charge combinations $m(K^+\pi^+)$ (bottom left) and $m(K^-\pi^-)$ (bottom right).

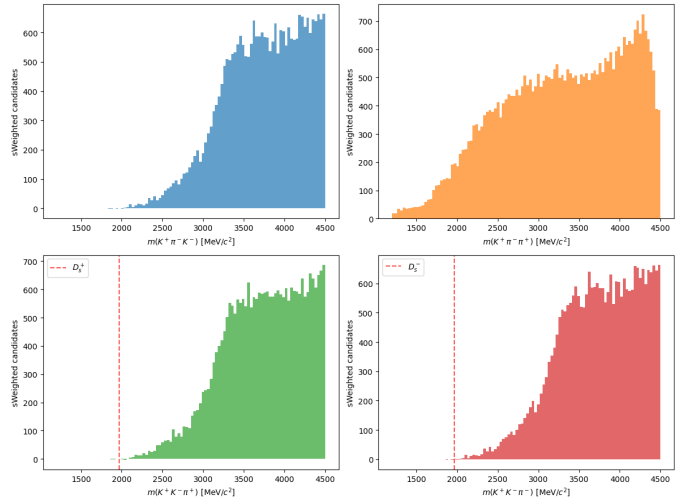

Fig. 19. sWeighted 3-body invariant mass distributions: $m(K^+\pi^-K^-)$ (top left), $m(K^+\pi^-\pi^+)$ (top right), $m(K^+K^-\pi^+)$ (bottom left) and $m(K^+K^-\pi^-)$ (bottom right)..

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